

Payload Field Map

VERSION	1.4
Date	23 September 2026
Author	Maria Azevedo
Status	Generated from delivered payloads; v1.1 added findings, pages, verification_status; v1.2 added confidence; v1.3 adds the engine's annual figure and calculation, rate_unit, payment_history, relationship in Cartograph's vocabulary and member_status; v1.4 adds the Work Number header to vois[] (overtime_rate, overtime_frequency, employment_start_date), drops the empty VOI row, and gives the questionnaire disclosure findings a category and subject

Every field the audit engine sends today, what each one means, and what it needs on the Cartograph side.

This replaces the ordering in **Schema Change Request v1.1**. Since v1.1 of this document the payload carries findings (section 8) and every income and asset record names the packet pages it was read from. What follows is what is arriving now.

Both delivered payloads are already in `raw_payload` and can be read alongside this document. This exists because the payload shows the shape and not the meaning, and several of the meanings decide compliance answers.

1. Six things that are not visible in the JSON

Read these before the tables. Each is a distinction the payload depends on and does not announce.

`current_value` **and** `manual_balance` **are not two amount columns**. `current_value` is a balance backed by a bank statement or a verification of assets. `manual_balance` is the resident's own figure with no third-party evidence. If they merge, a self-declaration satisfies a checklist item that exists to require verification. On both cases delivered so far, every asset arrived as `manual_balance`, none had supporting evidence.

`null` **is not** `false`. `is_disabled` and `full_time_student` are `true`, `false`, or `null`, and `null` means the packet never stated it. A default that turns `null` into `false` converts "unknown" into "not disabled", which is also one of the audit's own findings.

`ssn_last4` **is four digits by design**. Not truncated data. The engine holds full SSNs internally and masks them at every outbound point. Full digits are not sent because `CertHouseholdMember#ssn` has no encrypts declaration; that remains open in the schema request.

`ref` **and** `member_ref` **are scoped to one request**. They exist so income and asset records can point at a member inside a single payload. They carry no meaning outside it and should not be persisted as identifiers.

income_type or asset_type of other is a vocabulary gap, not a description. It means the extracted term matched nothing in the supplied picklist. The original value is named in the warnings, and those warnings are the list of terms still to agree.

Absent is not empty. Sections not listed here are omitted rather than sent empty, because an empty array asserts that nothing exists while absence asserts nothing at all. expense_records and findings are absent for that reason.

2. Envelope

Sent once at the top of every payload. These identify the case and the run that produced it.

FIELD	EXAMPLE	MEANING
schema_version	1.3	Payload contract version. Changes when the shape changes
extraction_id	ext_0ca98dfd...	Unique per run. A re-audit of the same case produces a new one, so it identifies the attempt rather than the case
engine_version	idp-1.0.0	Which build produced the extraction. Worth storing: it is what makes a result reproducible when a mapping question comes up months later
extracted_at	2026-09-07T17:06:12Z	UTC, ISO 8601. When extraction finished, not when the case was received
case_ref	J-PORT-05754	Your Job#ref_number, echoed back . This is the field to resolve against
target.job_id	5754	Echoed from the notification. Null if it was not sent
target.community_id	67	Echoed from the notification
target.unit_number	3-207	Echoed from the notification, and separately extracted onto cert_review from the certification itself. The two can disagree, which is worth surfacing rather than reconciling silently

Everything in target is echoed from what the notification carried, not read from the documents. cert_review.unit_number is the extracted one.

3. cert_review

One per case.

FIELD	EXAMPLE	MEANING
cert_type	annual	One of initial, annual, ar_self_cert, interim. Omitted entirely if the type cannot be determined — see the AR-SC note in section 7
effective_date	2026-10-01	ISO. Certification effective date
unit_number	3-207	As printed on the certification, which may differ from the unit record
hh_size	1	Count of members sent, not a figure read off the form
annual_income	42180.00	Household total as the certification declares it
annual_assets	915.00	Summed from the asset records sent, since the form carries no total
head_of_household_name	Dolores Ackerman	Convenience field; the authoritative flag is is_hoh on the member
tenant_rent	1180.00	
gross_rent	1245.00	
utility_allowance	65.00	
max_program_rent	1245.00	Program rent limit. Null when not extracted, which disables the rent-limit checks

All amounts are plain decimal strings: no currency symbol, no thousands separator, two decimal places.

4. household_members

FIELD	EXAMPLE	MEANING
ref	m01	Request-scoped handle. Do not persist
first_name	Dolores	
middle_name	R.	Often absent
last_name	Ackerman	
date_of_birth	1987-05-19	ISO. Null when the printed date could not be parsed, and reported in warnings
is_hoh	true	Exactly one member carries true

FIELD	EXAMPLE	MEANING
relationship	Minor Child	In Cartograph's vocabulary: Head of Household, Spouse, Co-Head, Minor Child, Other Adult, Live-in Aide, Foster Child, Unborn. The form's word (Dependent, Son, a 50059 code) is mapped by word and by age at the effective date: a child relationship under eighteen is Minor Child, from eighteen Other Adult. A word with no equivalent is sent as printed with a warning
relationship_as_printed	Dependent	The form's own word, unmapped
member_status	Active	Active, or Unborn for an expected child listed on the certification: no name, date of birth or SSN, counted in household size
is_disabled	null	true / false / null . See section 1
full_time_student	false	Same three-state rule
ssn_last4	4417	Four digits. See section 1
email	null	
cell_phone	555-0164	As printed; not normalized
sort_order	1	Order on the certification

5. income_records

One per income source. Paystubs arrive nested underneath rather than as a flat list, because grouping by member and employer happens engine-side.

FIELD	EXAMPLE	MEANING
member_ref	m01	Null when the earner could not be matched to a member, with a warning. The record is still sent, the income is real even when its owner is uncertain
income_type	wages_and_salaries	From the supplied picklist. other means no match; see section 1
source_name	Redwood Facilities Group LLC	Employer or payer as extracted

FIELD	EXAMPLE	MEANING
frequency_of_pay	bi-weekly	How often the person is paid. As the record states it; when it does not, the rate's own unit (a monthly benefit is paid monthly). Null for an hourly rate with no stated pay period
rate_unit	hourly	What rate_of_pay is per: hourly, daily, weekly, bi-weekly, semi-monthly, monthly, quarterly, annually, per_period. Null when the record has no rate
annual_income	29133.00	The engine's annual figure for this source. Null when no method could produce one (a record with no amount, a declaration with no period)
calculation	{method, annual_income, details, alternatives[]}	How the figure was reached. method is paystub-based, history-based, voi-based or self-declared; details is the arithmetic in words (avg(2 stubs) = 560.25 × 52 = 29133.00 – only 2 pay stub(s) in the file). alternatives[] are the other rows the engine computed for the source, each with status audit (a year-to-date projection run beside the primary), rejected (a method that produced an implausible figure, with why) or historical (income from a period before the certification). The methods-disagree findings are made from this comparison
payment_history	[{date, amount}]	For child support, alimony and benefit payment records: every payment line the document prints, as printed. date is null when the line prints no date (a numbered worksheet). The engine annualises from these rows; it never sums them on the document's behalf
date_received	2026-08-04	ISO. When the verification was received
employment_start_date	null	ISO
employment_status	Active	Active, Terminated, On Leave, or null. Needs a column
termination_date	null	ISO. Needs a column

FIELD	EXAMPLE	MEANING
self_declared_amount	812.50	The resident's own figure, distinct from anything verified
source_of_declaration	Application	Where the self-declared figure came from
verification_status	verified	verified (read from a third-party document), declared_only (the household declared it and nothing in the packet backs it), verified_not_declared (a document carries it and the certification does not), self_certified (AR-SC)
pages	[20, 21]	Packet pages this record was read from. Positions in the file Cartograph sent; a reviewer opens the same file and lands on the page
confidence	{score, flag, review[]}	The engine's confidence in this record: score 0 to 1, flag green / yellow / red, and review, the fields a reviewer should look at with field, flag and reason. See section 8a
paystubs[]		pay_date, gross_pay, ytd_amount, pay_frequency, pages
vois[]		voi_type, date_received, rate_of_pay, rate_unit, hours_per_pay_period, frequency_of_pay, ytd_amount, ytd_start_date, ytd_end_date, overtime_rate, overtime_frequency, employment_start_date. Sent only when the verification stated something (a rate, hours, a YTD figure or an overtime rate); a source verified by its stubs alone has no VOI row, so nothing arrives as a placeholder. frequency_of_pay here is the pay frequency, read from the stubs first. A Work Number report fills both: its pay period rows are the paystubs[], its header (rate, hours, start date, YTD as of a date, status) is the VOI row
zero_income	null	Object when a zero-income affidavit was filed

The annual_income on each record and the annual_income on cert_review answer different questions: the record's is what the engine computed from the documents, the review's is what the certification declares. When

they differ the CERT_SUMMARY_INCOME_MISMATCH finding says by how much.

A source can arrive with paystubs and no verification entry. That is not an error: when an employer verification comes back blank, the manager substitutes paystubs, and the engine reconstructs the source from them. Those records carry no self_declared_amount and no verification fields, because there is no third-party document behind them.

6. asset_records

FIELD	EXAMPLE	MEANING
member_ref	m01	Null when the owner could not be matched, with a warning
asset_type	checking	From the supplied picklist. other means no match
institution_name	Meridian Savings Bank	Bank or source
current_value	4830.00	Verified balance. Present only with a statement or a VOA behind it. See section 1
manual_balance	915.00	Self-declared balance
bank_stmt_avg_balance	null	Six-month average where stated
annual_income_from_assets	0.00	
interest_type	Percentage	Needs a column
percentage_of_ownership	100	Joint ownership share
source_of_declaration	null	
verification_status	verified	Same values as on income records
pages	[13]	Packet pages this record was read from
confidence	{score, flag, review[]}	As on income records
bank_statements[]		statement_date, balance. Only statements carrying at least one of the two are sent
voa	null	voa_date, reported_value, source

7. What needs adding

Three columns are being sent right now with nowhere to land:

TABLE	COLUMN	TYPE
cert_income_records	employment_status	string
cert_income_records	termination_date	date
cert_asset_records	interest_type	string

Two decisions, neither blocking:

AR-SC — this one does block. CertReview::CERT_TYPES allows `initial`, `annual`, `interim`. The engine audits four types, and AR-SC now arrives as `ar_self_cert`, a fourth value. It is not collapsed onto `annual`: AR-SC applies a different rule set — the certification form is the source of truth and no third-party wage verification is expected — and sending it as an ordinary annual produces false findings about documents the file is not supposed to have. CERT_TYPES needs the fourth value before an AR-SC case can import.

An unmappable or missing type is now **omitted** rather than defaulted. `cert_type` selects the checklist template through `cert_type_scope`, so a guessed value produces a clean-looking audit against the wrong rule set. A missing one is visible; a plausible wrong one is not.

source **on the cert child tables.** Re-import clears extraction-owned records and rewrites them. Without a marker separating what the importer created from what an analyst typed, the delete removes both. This only matters on the second import of a case, by which point reviewer work is in place. Until it exists, treat a cert review that already has records as a conflict rather than proceeding.

8. findings

One object per finding of the audit, in the shape of the Scan Findings thread. Field-level review notes ("Review recommended" on one field of one record) are the scorer's commentary, not findings, and are not sent.

FIELD	EXAMPLE	MEANING
finding_key	ASSET_SELF_DECLARED_VS_VERIFIED:randy_buck:savings	Stable identity: code plus the subject. A re-scan that reports the same finding sends the same key, so it updates the row rather than opening another
code	SIGNATURE_DATE_MISSING	The finding type. NOTE marks a finding that is still plain text; its key is a hash of the wording with digits removed, so a re-read figure updates the same row
category	income	file_review, income, asset, household_member, unit_rent
result	non_compliant	non_compliant is an issue to review. compliant and na are informational (a name spelled two ways, an amount that is a subtotal) and should not ask for a verdict

FIELD	EXAMPLE	MEANING
subject_type	income_record	household_member, income_record, asset_record, document, or null for the case
subject_label	Randy Buck – Midstates Bank	Human-readable subject
member_ref	m01	The member the finding is about, when it names one
pages	[13]	Where it was found. Empty when the finding is about the case as a whole (a missing form)
label	Self-declared asset balance differs from the verified balance	Short title
description		The full wording
correction_required		What resolves it, when the rule knows
assignment	client	internal, client, procedural_issue, or null
resolution_type	recalculation	presence_only or recalculation
disputes_extraction	true	The finding says the extraction contradicts the packet; these lower the engine's own confidence

Every scan sends the complete set. A key present in an earlier scan and absent from the latest one is cleared, not still open.

Category and subject_type are the key for the next action. A finding whose category is asset and whose subject is asset_record with no subject_label is a record that should exist and does not; its correction_required says so ("Add an asset record for the disclosed life insurance ..."). The questionnaire disclosure findings arrive this way as of 23 September: QUESTIONNAIRE_LIFE_INSURANCE_UNVERIFIED, _REAL_ESTATE_, _CHECKING_ and _SAVINGS_ as asset / asset_record; _EMPLOYMENT_, _SSA_, _CHILD_SUPPORT_ and _PENSION_ as income / income_record; QUESTIONNAIRE_STUDENT_UNVERIFIED as file_review with no subject, since the missing item is a certification and not a record. Each is assigned to the client and keyed per case, so a re-scan updates the row.

Verdicts come back on a separate findings_feedback event, nightly, one per case: case_ref, scan_id, then verdicts[] of finding_key, verdict (valid / invalid), verdict_reason, and manual_findings[] of description, page, subject_label, source: "manual", matched_checklist_item (key and name). The import-result callback is not the place for them: it arrives once at import time and a later call on it would overwrite the import outcome.

expense_records is still not sent. The engine does not extract expenses.

8a. confidence

On every household member, income record and asset record, on cert_review, and once at the top level for the case.

FIELD	EXAMPLE	MEANING
score	0.79	0 to 1. Green is 0.80 and above, yellow 0.50 and above, red below. A value never found on a source page is capped at 0.79, so green means "found in the document"
flag	yellow	What a reviewer triages by. A record the household only declared, or one a finding disputes, cannot be green
review	<pre>[{"field": "rateOfPay", "flag": "yellow", "reason": "Found only on the certification form, not in this record's documents"}]</pre>	The fields to look at, each with the reason. Empty on a green record

The top-level object carries score, flag and the field counts (green, yellow, red, na). The case flag is bounded by the worst disputed record and is yellow whenever any income is declared-only; it is a triage signal, not an average.

Green fields can go straight in. Yellow and red are the review list; the reason says what to check.

9. On the warnings

Every payload carries a list of what the mapping could not carry: a date that would not parse, an earner who matched no member, a type with no equivalent in the picklist. They are the only record of where the two vocabularies disagree, and the accumulated list is the agenda for closing the gaps.

The same applies in the other direction. Please populate warnings on the import result for every value that collapses, every date that fails to parse, and every member that arrives without a relationship, without them, that information is recoverable only by manual spot-checking.